

Introduction to Transfer Learning

Contents

- ✓ What is Transfer Learning
- ✓ How Does Transfer Learning Work
- ✓ Approaches to Transfer Learning
- ✓ Advantages of Transfer Learning
- ✓ Applications of Transfer Learning in Computer Vision

What is Transfer Learning

A machine learning technique where a model trained on one task is reused as the foundation for a second task

Why do we need Transfer learning?

Limited Data



Many real-world tasks don't have large labeled **datasets** (e.g., medical images, satellite data).

Enhanced Performance



Models trained from scratch on small data often overfit.

Time and Cost Efficiency



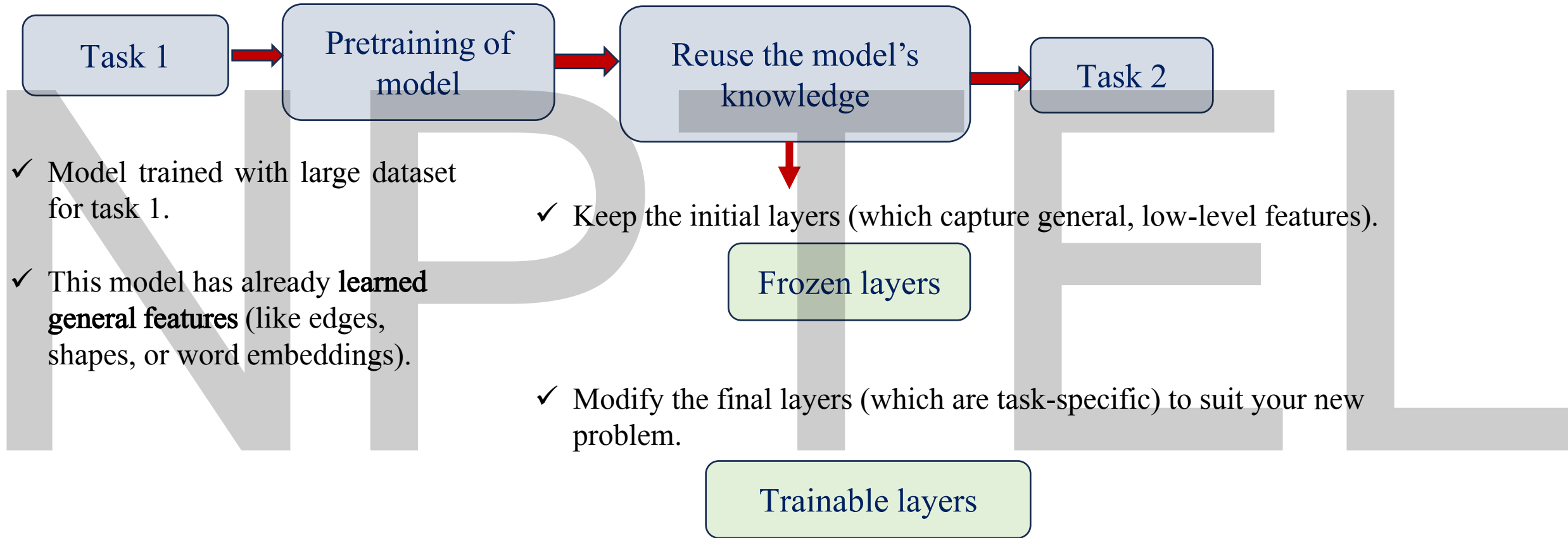
Training deep neural networks from scratch can take days or weeks.

Adaptability



A model trained in one domain (e.g., natural images) can be adapted to another (e.g., medical X-rays).

How Does Transfer Learning Work



Frozen and Trainable Layers in Transfer Learning

Frozen Layer

- Layers whose parameters (weights & biases) are **not updated** during training on the new dataset.
- Preserve previously learned features (like edges, textures, shapes in vision models or grammar in NLP models).

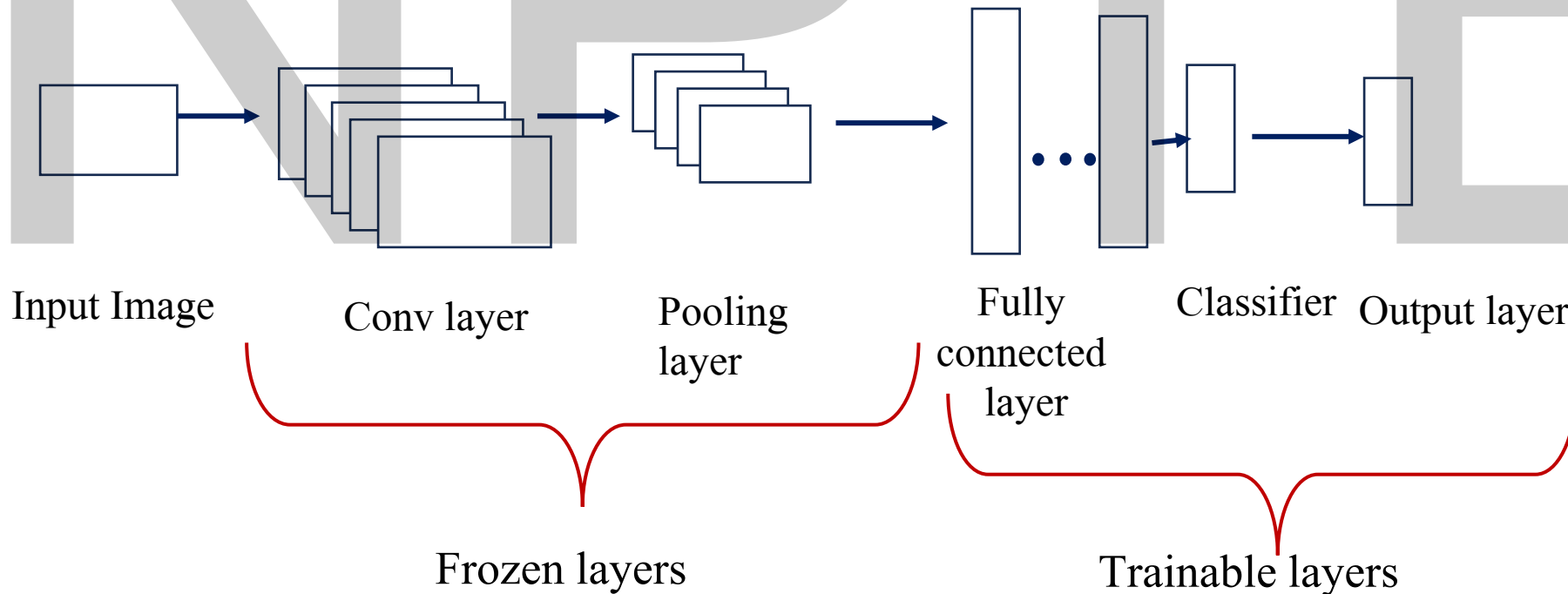
Trainable Layer

- Layers whose parameters are **updated** (through backpropagation) during training on the new dataset.
- Allow the model to adapt and specialize to the new task/domain.
- Capture dataset-specific patterns that pre-trained layers may not have learned.

Approaches to Transfer Learning

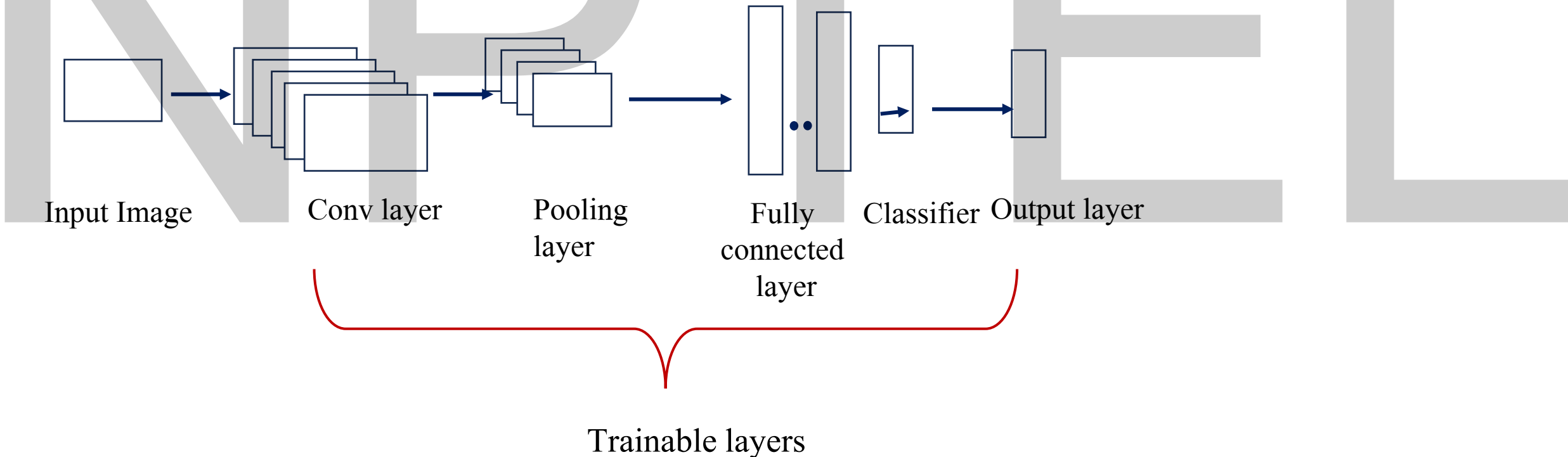
Feature Extraction

- Uses the representations learned by the pre-trained network to extract meaningful features for new samples of a specific task
- Freeze all layers of the pre-trained model
- Only train a new classifier (like softmax or dense layer) on top of it.
- Useful when the target task has a small dataset or when computational resources are limited



Fine Tuning

- pre-trained model's parameters are further trained (fine-tuned) on the target task's dataset along with the additional layers.
- Unfreeze some of the layers of the pretrained/base model.
- Retrain them on the new dataset with a smaller learning rate
- Allows the model to adapt its knowledge to the new domain.
- It is typically used when the target task dataset is relatively large and similar to the pre-trained model dataset



Advantages of Transfer Learning

- ✓ Reduces training time as it start with pre-learned knowledge instead of training from scratch.
- ✓ Requires less data and hence useful when the target task has limited labeled data (in medicine, defense, etc).
- ✓ Pre-trained features usually generalize well, giving better accuracy compared to models trained from scratch, thus improving performance.
- ✓ Saves computation power and storage.

Applications of Transfer Learning

Computer Vision

Popular Pretrained Models in Computer Vision includes

- VGG (Visual Geometry Group) models (e.g., VGG16, VGG19)
- ResNet (Residual Networks) models (e.g., ResNet-50, ResNet-101)
- Inception (GoogLeNet) models (e.g., Inception v3)
- MobileNet models (e.g., MobileNetV2, MobileNetV3)
- EfficientNet
- DenseNet (Dense Convolutional Network)
- YOLO (You Only Look Once)

They are used in:

Image Classification

Object Detection
& Recognition

Image
Segmentation

Face Recognition

Medical Imaging

Few Shot and Zero Shot Learning

Few-shot and zero-shot learning are specific applications of transfer learning, where the model transfers knowledge from pre-training to new tasks with very limited or no labeled data.

Few-Shot Learning



The model learns to perform a new task with only a few labeled examples per class.

Zero-Shot Learning



The model performs a task without seeing any labeled examples of that task during training.

NPTTEL

Thank you